ELEC50011 PART B: PROBABILITY AND STATISTICS – SOLUTIONS

1. Consider two continuous random variables X and Y that have the joint probability density function (pdf) given in the following equation:

$$f_{X,Y}(x,y) = \begin{cases} ae^{x+y}, & \text{if } x>0, y>0 \text{ and } 0<x+y<1, \\ 0, & \text{otherwise.} \end{cases}$$

Here, $a>0$ is a constant.

a) Find the value for a that ensures that $f_{X,Y}(x,y)$ describes a valid pdf. [4]

SOLUTION. Since $a>0$, it is already ensured that $f_{X,Y}(x,y) \geq 0$. The remaining requirement is that: [1]

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{X,Y}(x,y)dxdy = 1$$

Plugging in the expression for $f_{X,Y}(x,y)$: [2]

$$\begin{aligned}
\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{X,Y}(x,y)dxdy &= c\int_0^1\int_0^{1-y} e^{x+y}dxdy \\
&= a\int_0^1\int_0^{1-y} e^x e^y dxdy \\
&= a\int_0^1 e^y \left[e^x\right]_0^{1-y} dy \\
&= a\int_0^1 e^y \left(e^{1-y}-e^0\right) dy \\
&= a\int_0^1 (e-e^y)dy \\
&= a\left[ey-e^y\right]_0^1 \\
&= a\left[e-e-0+1\right]_0^1 \\
&= a
\end{aligned}$$

Hence, $a=1$ is needed to ensure $f_{X,Y}(x,y)$ is a valid pdf. [1]

b) Are X and Y independent? Justify your answer. [3]

SOLUTION. X and Y are not independent. [1]

This can be verified manually by evaluating the marginals, but it is not necessary. The joint pdf cannot be factorize due to the coupling constraint that $x+y<1$. [2]

c) Find the marginal $f_X(x)$. [5]

SOLUTION. We integrate the joint pdf over y to find the marginal of X. [1]

Working out the integral for any $0 < x < 1$: [4]

$$\begin{aligned} f_X(x) &= \int_0^{1-x} f_{X,Y}(x,y)dy \\ &= \int_0^{1-x} e^{x+y}dy \\ &= \int_0^{1-x} e^x e^y dy \\ &= e^x \left[e^y\right]_0^{1-x} \\ &= e^x \left(e^{1-x} - 1\right) \\ &= e - e^x \end{aligned}$$

Note that by symmetry it follows that $f_Y(y) = e - e^y$, and we can immediately see that X and Y are not independent:

$$f_X(x) \times f_Y(y) = (e - e^x)(e - e^y) \neq e^{x+y} = f_{X,Y}(x,y)$$

This should provide students who were unsure about their answer to part (b) with some reassurance.

d) Compute the expected values of X and Y, i.e., $E(X)$ and $E(Y)$. [3]

SOLUTION. By definition: [1]

$$E(X) = \int_{-\infty}^{\infty} x f_X(x)dx$$

Plugging in the marginal and applying integration by parts: [2]

$$\begin{aligned} E(X) &= \int_0^1 x(e - e^x)\,dx \\ &= e\int_0^1 x dx - \int_0^1 x e^x dx \\ &= e\left[\frac{1}{2}x^2\right]_0^1 - \left[xe^x\right]_0^1 + \int_0^1 e^x dx \\ &= e\left[\frac{1}{2}x^2\right]_0^1 - \left[xe^x\right]_0^1 + \left[e^x\right]_0^1 \\ &= \frac{e}{2} - 0 - e + 0 + e - 1 \\ &= \frac{e}{2} - 1 \end{aligned}$$

e) Compute the variance of X, i.e., $Var(X)$. [3]

SOLUTION. We begin with: [2]

$$\begin{aligned} E(X^2) &= \int_0^1 x^2(e - e^x)\,dx \\ &= e\int_0^1 x^2 dx - \int_0^1 x^2 e^x dx \\ &= e\left[\frac{1}{3}x^3\right]_0^1 - \left[x^2e^x\right]_0^1 + 2\int_0^1 xe^x dx \\ &= \frac{e}{3} - 0 - e + 0 + 2 \\ &= 2 - \frac{2}{3}e \end{aligned}$$

To find the variance: [1]

$$Var(X) = E(X^2) - (E(X))^2 = 2 - \frac{2}{3}e - \left(\frac{e}{2} - 1\right)^2$$

f) Find the covariance between X and Y, i.e., $Cov(X,Y)$. [4]

SOLUTION. We first look for: [3]

$$\begin{aligned}
E(XY) &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xy f_{X,Y}(x,y)dxdy \\
&= \int_0^1 \int_0^{1-y} xye^{x+y}dxdy \\
&= \int_0^1 ye^y \int_0^{1-y} xe^x dxdy \\
&= \int_0^1 ye^y \left[xe^x - e^x\right]_0^{1-y} dy \\
&= \int_0^1 ye^y \left((1-y)e^{1-y} - e^{1-y} + 1\right) dy \\
&= \int_0^1 ye^y \left(1 - ye^{1-y}\right) dy \\
&= \int_0^1 ye^y dy - e\int_0^1 y^2 dy \\
&= 1 - \frac{e}{3}
\end{aligned}$$

To find the covariance, we then compute: [1]

$$Cov(X,Y) = E(XY) - E(X)E(Y) = 1 - \frac{e}{3} - \left(\frac{e}{2} - 1\right)^2$$

g) Find the conditional probability density function of Y given $X = x$. [2]

SOLUTION. By definition: [1]

$$f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)}$$

Plugging in the expressions we have obtained so far: [1]

$$f_{Y|X}(y|x) = \frac{e^{x+y}}{e - e^x}$$

under the condition $0 < y < 1 - x$, and 0 otherwise. Let us verify that this is indeed a valid pdf to check our work (this is not necessary in the exam):

$$\int_0^{1-x} \frac{e^{x+y}}{e-e^x} dy = \frac{e^x}{e-e^x}\int_0^{1-x} e^y dy = \frac{e^x}{e-e^x}\left(e^{1-x} - 1\right) = 1$$

h) Compute the conditional expectation of Y given $X = x$. [3]

SOLUTION. By definition: [1]

$$E(Y|X=x) = \int_{-\infty}^{\infty} y f_{Y|X}(y|x)dy = \frac{e^x}{e-e^x}\int_0^1 ye^y dy$$

Plugging in what we have found so far and evaluating the expressions: [2]

$$\begin{aligned} E(Y|X=x) &= \frac{e^x}{e-e^x}\int_0^{1-x} ye^y dy \\ &= \frac{e^x}{e-e^x}\left[ye^y - e^y\right]_0^{1-x} \\ &= \frac{e^x}{e-e^x}\left((1-x)e^{1-x} - e^{1-x} + 1\right) \\ &= \frac{e^x}{e-e^x}\left(1 - xe^{1-x}\right) \\ &= \frac{e^x - ex}{e-e^x} \end{aligned}$$

i) Evaluate $E(E(Y|X))$ and compare to $E(X)$ and $E(Y)$. How and why do these relate? [3]

SOLUTION. We have: [1]

$$E(E(Y|X)) = \int_0^1 \frac{e^x - ex}{e-e^x}(e-e^x)\,dx = \int_0^1 e^x dx - e\int_0^1 x dx = e - 1 - \frac{e}{2} = \frac{e}{2} - 1$$

From the tower-rule of expectation, we know that $E(E(Y|X)) = E(Y)$. [1]

Since X and Y are symmetric, we have $E(X) = E(Y)$ and hence all three quantities are identical. [1]

2. You are presented with a bag containing red balls and blue balls. The total number of balls is 100. Of the 100 balls, a fraction of p are red, where $0 < p < 1$ is unkown to you. This means that if you pick a random ball from the bag, the probability that it will be red is p. You would like to estimate p, and with this goal in mind you construct an experiment. You pick balls from the bag with replacement until you see a red ball for the first time, and record in the random variable X the number of tries it took.

a) Formulate an expression for the probability mass function of X and show that it is a valid pmf. What is the name of this distribution? [5]

SOLUTION. The event $\{X = x\}$ describes the situation where we saw exactly $x-1$ blue balls, followed by a red ball. It follows that: [2]

$$f_X(x) = P(X = x) = (1-p)^{x-1}p$$

To show that it is a valid pmf, we need to check whether it adds up to one. Indeed, by the geometric series: [2]

$$\sum_{x=1}^{\infty}(1-p)^{x-1}p = p\sum_{x=0}^{\infty}(1-p)^x = \frac{p}{p} = 1$$

This distribution is known as the geometric distribution. [1]

b) You repeat the experiment n times under identical conditions, collecting the outcome of the i-th experiment in X_i, resulting in an i.i.d. sample of size n, denoted by $X_1, X_2, \ldots, X_n$. Formulate the maximum likelihood estimate of p given $x_1, x_2, ..., x_n$ and show that it maximizes the likelihood. [6]

SOLUTION. We first formulate the likelihood: [1]

$$\mathscr{L}_X(p) = \prod_{i=1}^{n} f_X(x_i) = p^n \prod_{i=1}^{n}(1-p)^{x_i - 1}$$

As usual, it is convenient to maximize the log instead: [1]

$$\ell_X(p) = \log \mathscr{L}_X(p) = n \log p + \log(1-p) \sum_{i=1}^{n} (x_i - 1)$$

Taking derivatives and setting to zero: [1]

$$\frac{d}{dp} \ell_X(p) = \frac{n}{\hat{p}} - \frac{\sum_{i=1}^{n} (x_i - 1)}{1 - \hat{p}} = 0$$

After rearranging: [2]

$$\frac{1}{\hat{p}} - 1 = \frac{\sum_{i=1}^{n} (x_i - 1)}{n} \Longleftrightarrow \hat{p} = \frac{n}{\sum_{i=1}^{n} x_i} = \frac{1}{\bar{x}}$$

We verify that this is indeed a maximum via the second derivative: [1]

$$\frac{d^2}{dp^2} \ell_X(p) = -\frac{n}{p^2} - \frac{\sum_{i=1}^{n} (x_i - 1)}{(1-p)^2} < 0$$

c) Find the method-of-moments estimate of p based on $x_1, x_2, \ldots, x_n$. [4]

SOLUTION. We know that $E(X) = \frac{1}{p}$. [2]

Hence, for the method-of-moments method we have: [2]

$$p = \frac{1}{E(X)} \Longrightarrow \hat{p} = \frac{1}{\bar{x}} = \frac{1}{\frac{\sum_{i=1}^{n} x_i}{n}} = \frac{n}{\sum_{i=1}^{n} x_i}$$

d) Are the resulting estimators unbiased? [5]

SOLUTION. In both cases the estimator takes the form $\hat{p} = \frac{1}{\bar{x}}$. This estimator is biased. [2]

To verify this, we note that: [3]

$$E(\widehat{p}) = E\left(\frac{1}{\bar{x}}\right) \neq \frac{1}{E(\bar{x})} = \frac{1}{E(X)} = p$$